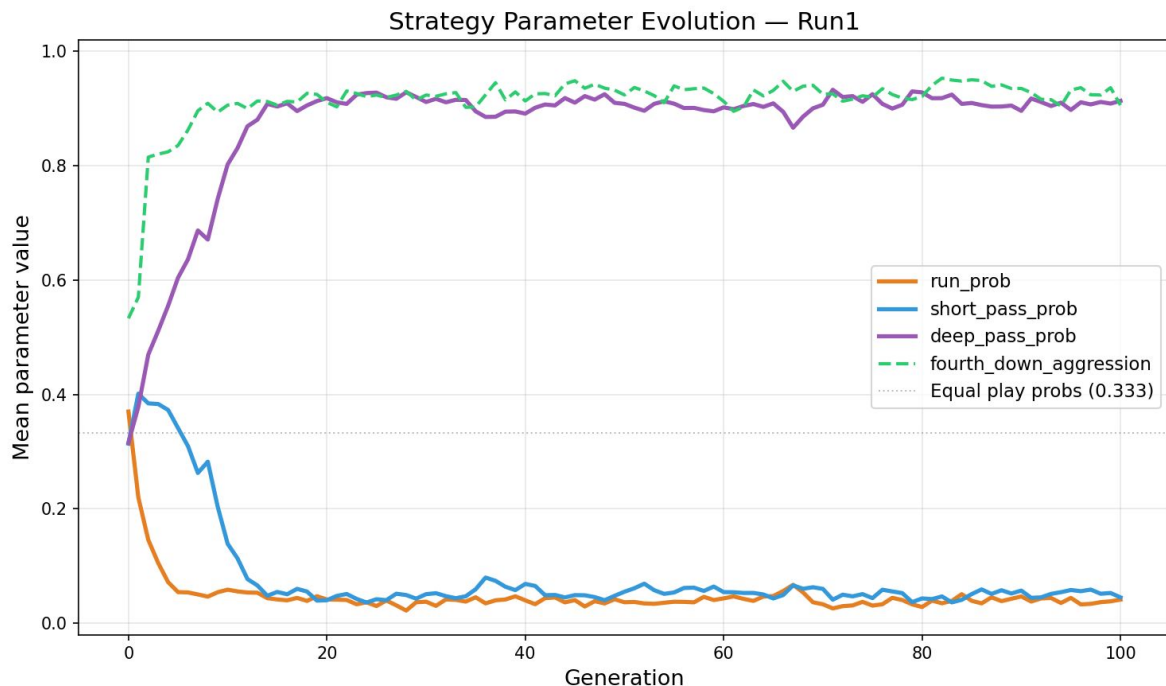


Evolving Offensive Football Strategies Using Genetic Algorithms

Jordan Martinez

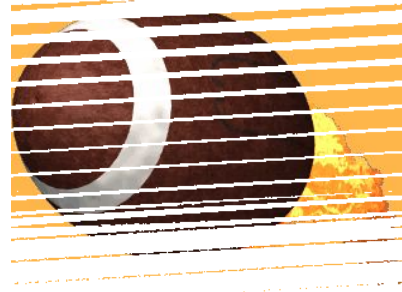


Question



- **Main Question:**
Does an offensive play-calling strategy emerge when using Genetic Algorithms? If so, what strategy emerges?
- How play calling is a CAS?
 - **Collective Behavior** — drive outcomes emerge from individual play interactions
 - **Emergence** — optimal strategies arise without explicit programming
 - **Adaptation** — play-calling evolves in response to scoring feedback

Methods



- **Data:** 106,000+ NFL plays (2022–2024) via nflfastR
- **Drive Simulator:** tracks down, yards to go, and field position across a full offensive possession. Calculates given score.
- **Fitness Function:** $f = \text{mean}[\text{points}/\text{drive}]$
 - refined across experiments to $f = \text{mean} - 0.5 \times \text{std}$ to penalize inconsistent high-variance strategies
- **GA Parameters** [each strategy is encoded as a genome: run%, short pass%, deep pass%, and fourth down aggression]:
 - **Population:** 100 strategies, 100 generations, 10 random seeds
 - **Tournament Selection:** 5 random strategies compete, fittest becomes parent
 - **Uniform Crossover:** each of 4 genome parameters independently inherited from either parent, renormalized
 - **Gaussian Mutation:** noise added [mean=0, std=0.05], clipped and renormalized
 - **Elitism:** top 5 strategies preserved each generation

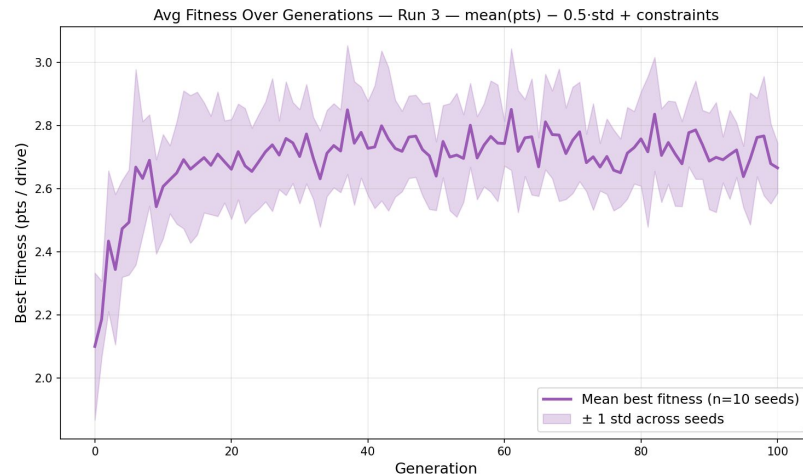


Results



Run 1 [Baseline]

- **Key Difference:** Raw mean fitness, no penalties, no constraints
- **Play-call Split:** Deep 96.6%, Short 1.6%, Run 1.8%
- **4th Down Aggression:** 0.963
- **Fitness:** 4.317 ± 0.144 pts/drive



Run 3 [Variance Penalty + Situational Constraints]

- **Key Difference:** fitness = mean - 0.5×std, deep pass capped in short yardage and own territory, short pass boosted on long downs
- **Play-call split:** Deep 91.2%, Short 5.5%, Run 3.3%
- **4th Down Aggression:** 0.953
- **Fitness:** 2.666 ± 0.083 pts/drive

Results



Run 4a [Turnover Penalty, Raw Fitness]

- **Key Difference:** fitness = mean[pts], -3.5 point penalty applied on every turnover with Run 3 constraints
- **Play-call split:** Short 75.4%, Deep 21.6%, Run 3.0%
- **4th Down Aggression:** 0.930
- **Fitness:** 3.563 ± 0.234 pts/drive

Key Finding and Future Work

- **Key Finding**

- Emergence occurs consistently across all runs, but the emergent strategy is determined entirely by the fitness landscape such as penalizing turnovers, which causes the GA to invert from deep-pass dominant to short-pass dominant.

- **Future Work**

- **Run 5** — field position adjusted EPA values for punts and failed 4th down conversions, reflecting that failing a 4th down in own territory is significantly more costly than punting from the same position
- **Score differential awareness** — adjusting fourth down aggression based on whether the team is trailing or leading
- **Co-evolutionary defense genome** — adding a defensive agent that adapts against the offense

Questions :]